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SAM 3D Animal: Promptable Animal 3D Reconstruction from Images in the Wild

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Figure 1: A promptable view of multi-animal 3D reconstruction. We present **SAM 3D Animal**, a promptable framework that addresses the problem of joint 3D reconstruction of multiple animals from a single image. Here, we show the input image together with the overlay reconstruction results.

Abstract

3D animal reconstruction in the wild remains challenging due to large species variation, frequent occlusions, and the prevalence of multi-animal scenes, while existing methods predominantly focus on single-animal settings. We present SAM 3D Animal, the first promptable framework for multi-animal 3D reconstruction from a single image. Built on the SMAL+ parametric animal model, our method jointly reconstructs multiple instances and supports flexible prompts in the form of keypoints and masks which enable more reliable disambiguation in crowded and occluded scenes. To train such a model, we further introduce Herd3D, a multi-animal 3D dataset containing over 5K images, designed to increase diversity in species, interactions, and occlusion patterns. Experiments on the Animal3D, APTv2, and Animal Kingdom datasets show that our framework achieves state-of-the-art results over both existing model-based and model-free methods, demonstrating a scalable and effective solution for prompt-driven animal 3D reconstruction in the wild. Project page: `https://georgehux.com/SAM3D-Animal-project-page/`

1 Introduction

Animals are a fundamental part of the visual world, yet 3D reconstruction research remains heavily skewed toward humans. Human-centric methods have advanced pose and shape estimation dramatically [21, 61, 60, 18, 47, 6, 29]. In contrast, animal reconstruction still suffers from scarce datasets, broad species variation, and inconsistent anatomical definitions.

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Parametric animal models such as SMAL [65] and SMAL+ [63] provide an effective basis for recovering 3D pose and shape from a single image [64, 53, 39, 36, 2]. These approaches typically focus on one animal at a time and often rely on pre-cropped inputs or strong object detections. However, many in-the-wild animal scenes contain multiple individuals with mutual occlusion, and complex interactions that invalidate single-animal assumptions.

Multi-animal 3D reconstruction raises unique challenges beyond those of the single-object case. First, instance association becomes ambiguous when animals overlap or occlude one another. Second, pose and shape estimation must be jointly consistent across multiple hypotheses, since mistakes on one individual can be amplified by false depth ordering or incorrect occlusion reasoning. Third, available datasets rarely provide dense multi-animal 3D annotations, which hinders supervised learning for crowded scenes.

To overcome these challenges, we draw inspiration from promptable reconstruction in human vision. Recent works such as SAM 3D Body [56] demonstrates that explicit prompts can guide a model to focus on a desired subject and resolve ambiguity in cluttered scenes. Prompts can take the form of keypoints or masks, each providing a different level of spatial and semantic guidance.

In this paper, we introduce **SAM 3D Animal**, the first promptable framework for multi-animal 3D reconstruction, see Fig. 1. Our model uses the SMAL+ template [63] and can ingest optional prompts in two modalities: keypoints for skeletal alignment and masks for precise silhouette discrimination. This promptable design allows SAM 3D Animal to recover multiple animals jointly from a single image. Different from SAM 3D Body, which reconstructs a single prompted subject per forward pass, our model adopts a set-prediction paradigm that recovers all animal instances in one shot via DETR-style [12] bipartite matching, eliminating the need for per-instance bounding-box cropping.

However, training such a multi-instance model with only 2D-annotated datasets is insufficient, as 2D keypoints and silhouettes alone cannot provide the per-instance 3D shape supervision needed to resolve inter-animal occlusions. To address this, we propose Herd3D, a multi-animal 3D dataset containing over 5K images with per-instance ground-truth meshes, designed to increase diversity in species, interactions, and occlusion patterns. The generation pipeline of Herd3D is adapted from GenZoo [39], where each animal is synthetically generated to produce an image with an image-aligned SMAL+ model.

To demonstrate the effectiveness of SAM 3D Animal, we compare it with state-of-the-art animal mesh recovery methods on publicly available datasets including Animal3D [53], APTv2 [57], and Animal Kingdom [38]. Even without any prompt, our model achieves competitive or superior results compared to existing methods. When prompts are provided, performance improves consistently across all benchmarks, with up to 54% AP gain and 80% mAP gain on the out-of-domain Animal Kingdom dataset over the strongest baseline, as well as 5.2 PA-MPJPE improvement on Animal3D. Ablation studies confirm that Herd3D brings consistent improvements, particularly on multi-animal benchmarks, and that keypoint prompts are the dominant contributor among prompt modalities, with performance scaling monotonically as the number of keypoints increases.

2 Related Work

2.1 Model-Free Reconstruction

Model-free animal reconstruction learns 3D structure directly from image or video collections without assuming a predefined template [17, 52]. Early methods model category-specific articulated animals from single-view image collections by separating a predefined skeleton prior from instance-specific deformations [59, 51, 50, 19]. Later approaches extend to a wider variety of species, either by learning a unified shape model [30] or by applying linear skinning to deform learned 3D object shapes [4]. However, these methods still struggle with extreme poses, heavy occlusions, and limited viewpoint coverage, often producing geometrically ambiguous reconstructions.

2.2 Model-Based Reconstruction

Model-based animal reconstruction typically relies on predefined quadruped templates such as SMAL [65] that encode the shape and articulation structure of specific animal families. These models either fit predefined 3D templates to animal images using 2D observations such as keypoints or

silhouettes [64, 8, 9], or directly reconstruct the 3D shape from image or video observations [13, 55, 59, 66, 35, 14]. This parametric formulation offers interpretable and controllable representations, which make the reconstructed animals readily animatable and editable. Recent works further extend SMAL-based reconstruction to broader quadruped species and training settings. AWOL [63] maps CLIP-style embeddings to the SMAL$^+$ parameter space for language- and image-guided animal shape generation. RAW [24] reconstructs animals jointly with their surrounding environment, including multi-animal scenes. However, it relies on rigid animal assets rather than articulated animal models, and therefore does not address fine-grained articulated animal reconstruction.

2.3 Animal Pose Estimation Datasets

In comparison to humans, the construction of large-scale animal datasets is significantly more challenging because animals are difficult to capture in controlled environments and exhibit substantial morphological diversity across species. Animal benchmarks, such as Stanford Extra [7], Animal Pose [10], and AwA-Pose [5], remain limited to 2D annotations. Existing 3D animal datasets, such as Animal3D [53] and GenZoo [39] predominantly focus on single-animal instances. Additionally, multi-view quadrupedal animal datasets focus mostly on single species such as dogs [41, 48, 22], horses [26], and some agriculture animals like pigs [3], goats [20] or cattle [40]. Meanwhile, real-captured dynamic datasets targeting wildlife animals such as WildDepth [1] and AiM [62], and sim-to-real datasets for diverse species [35, 32, 37, 34, 27], all focus on single instances. Previous large-scale social animal benchmarks like APT-36K [58] and Animal Kingdom [38] provide only 2D annotations. This limitation restricts the development of methods that can jointly model inter-animal occlusions, spatial relationships, and pose dependencies in multi-animal scenes.

2.4 Promptable Mesh Reconstruction

Promptable mesh reconstruction has recently emerged in human mesh recovery, where auxiliary cues guide 3D estimation under occlusion and crowding. PromptHMR [46] incorporates full-image context with spatial and semantic prompts for pose and shape estimation. SAM 3D Body [56] extends this idea to full-body recovery through a promptable encoder-decoder architecture supporting keypoint and mask prompts. SAM-Body4D [16] further leverages temporally consistent masklets to produce coherent mesh trajectories from videos. However, these methods are designed for humans, whereas our work targets the animal domain, where large morphological diversity, inter-animal occlusion, and multi-instance interactions must be jointly considered.

3 SAM 3D Animal Model

3.1 Preliminary

SMAL+. SMAL+ [63], denoted as $\mathcal{M}(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\gamma})$, extends the original SMAL [65] model by incorporating training samples from D-SMAL [43] and hSMAL [25], alongside new species such as the giraffe, bear, mouse and rat. This results in a broader, 145-dimensional shape space learned from a total of 145 animals. The inputs to SMAL+ are the shape parameters $\boldsymbol{\beta} \in \mathbb{R}^{145}$ and the pose parameters $\boldsymbol{\theta} \in \mathbb{R}^{35\times 3}$ (using an axis-angle representation) and the global translation $\boldsymbol{\gamma} \in \mathbb{R}^3$. By applying linear blendshapes and Linear Blend Skinning (LBS), SMAL+ outputs a posed mesh with vertices $\mathbf{V} \in \mathbb{R}^{3889\times 3}$ and faces $\mathbf{F} \in \mathbb{N}^{7774\times 3}$.

3.2 End-to-end Multi-Instance Network

Given an animal image, our model can reconstruct all animals in the image without requiring preprocessed bounding boxes, and support masks or keypoints as prompts.

Encoder. Starting from the image $\boldsymbol{x} \in \mathbb{R}^{H\times W\times 3}$, we utilize the ViT-Huge Encoder [15] to generate the feature tokens $\mathcal{F} \in \mathbb{R}^{(H_0\times W_0)\times C_0}$, where C_0, H_0, and W_0 are the channels, the height and width of the feature map, respectively. In our case, $W = H = 512$, $C_0 = 1280$, $W_0 = H_0 = 32$.

Decoder. Inspired by SAM 3D Body [56], the decoder is a SAM-style promptable Transformer (see Fig. 2). Specifically, it takes the feature tokens $\mathcal{F}$ and a set of query tokens as input, then performs cross-attention, and finally predicts the SMAL+ parameters, cameras, and bounding boxes. Note

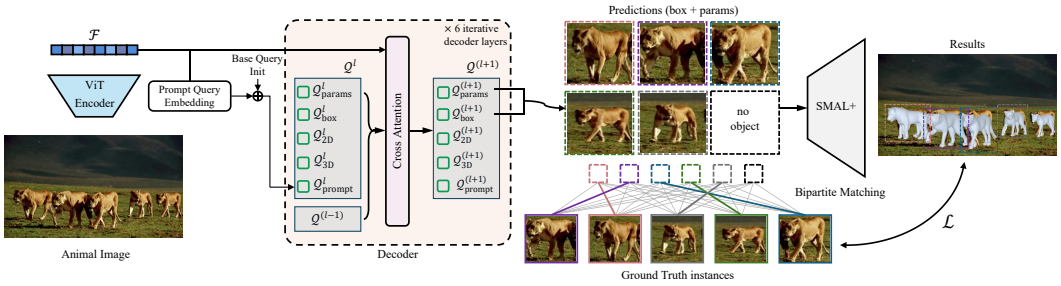


Figure 2: SAM 3D Animal model structure.

that, different from SAM 3D Body which predicts single person at a time, our model directly predict $P = 30$ possible instances at a time, eliminating the need for bounding box input. The query tokens consist of five distinct token groups at state index $l = 0, \ldots, L$, where $L = 6$ and $l = 0$ denotes the decoder input:

$$\mathcal{Q}^{(l)} = \text{Concat}(\mathcal{Q}^{(l)}_{\text{params}}, \mathcal{Q}^{(l)}_{\text{box}}, \mathcal{Q}^{(l)}_{\text{2D}}, \mathcal{Q}^{(l)}_{\text{3D}}, \mathcal{Q}^{(l)}_{\text{prompt}}) \in \mathbb{R}^{N \times D}, \tag{1}$$

where $\mathcal{Q}^{(l)}_{\text{params}}$, $\mathcal{Q}^{(l)}_{\text{box}}$, $\mathcal{Q}^{(l)}_{\text{2D}}$, and $\mathcal{Q}^{(l)}_{\text{3D}}$ represent the SMAL+ pose tokens, bounding box tokens, 2D keypoint tokens, and 3D keypoint tokens, respectively, while $\mathcal{Q}^{(l)}_{\text{prompt}}$ represents the interaction prompt tokens. The four base query groups, $\mathcal{Q}^{(0)}_{\text{params}}$, $\mathcal{Q}^{(0)}_{\text{box}}$, $\mathcal{Q}^{(0)}_{\text{2D}}$, and $\mathcal{Q}^{(0)}_{\text{3D}}$, are learnable embeddings. They are randomly initialized once before training and subsequently optimized jointly with the network. At inference time, their learned values are used without reinitialization. In contrast, $\mathcal{Q}^{(0)}_{\text{prompt}}$ is input-dependent and constructed from the keypoint prompts provided for the current image, rather than being a randomly initialized base query. A keypoint prompt $p_i = (u_i, v_i, c_i)$ comprises normalized coordinates $(u_i, v_i) \in [0, 1]^2$ and a label: an Animal3D keypoint ID 0–25, a negative point -1, or an invalid, missing, or padded point -2. Its token is

$$q^{(0)}_{\text{prompt},i} = W_p e_i + b_p, \qquad e_i = \begin{cases} \text{PE}(u_i, v_i) + E(c_i), & c_i \in \{0, \ldots, 25\}, \\ E_{\text{neg}}, & c_i = -1, \\ E_{\text{invalid}}, & c_i = -2, \end{cases}$$

where PE denotes positional encoding, $E(c_i), E_{\text{neg}}, E_{\text{invalid}}$ are learned label embeddings, and W_p, b_p are learned projection parameters. Prompts enter the first decoder layer through $\mathcal{Q}^{(0)} = \text{Concat}(\mathcal{Q}^{(0)}_{\text{base}}, \mathcal{Q}^{(0)}_{\text{prompt}})$, with $\mathcal{Q}^{(0)}_{\text{base}}$ collecting the four base query groups. The feature dimension is $D = 1024$. At inference, $N = 55P + N_{\text{prompt}} = 1650 + N_{\text{prompt}}$: each hypothesis has one parameter, one previous-estimate, one box, and 26 tokens each for 2D and 3D keypoints; N_{prompt} counts prompt tokens including padding.

Decoder block $l + 1$ maps $\mathcal{Q}^{(l)}$ to $\mathcal{Q}^{(l+1)}$ (Fig. 2):

$$\mathcal{Q}^{(l+1)} = \mathcal{D}_{l+1}(\mathcal{Q}^{(l)}, A^{(l)}, \mathcal{F}), \qquad l = 0, \ldots, L - 1, \tag{2}$$

Here, $\mathcal{D}_{l+1}$ comprises self-attention, image cross-attention, and a feed-forward network; $A^{(l)}$ denotes query positional augmentation. Fig. 2 highlights cross-attention. Its lower $\mathcal{Q}^{(l-1)}$ label schematically denotes previous-estimate conditioning, encoded once before decoding (using the initial estimate if none is supplied); it does not index a decoder layer.

For $l = 0, \ldots, L - 2$, SMAL+ and camera predictions yield projected 2D coordinates $\hat{x}^{(l+1)}_{\text{2D}}$ and pelvis-centered 3D coordinates $\bar{x}^{(l+1)}_{\text{3D}}$. Feedback updates the next input in place:

$$\begin{aligned} \mathcal{Q}^{(l+1)}_{\text{2D}} &\leftarrow \mathcal{Q}^{(l+1)}_{\text{2D}} + \phi_{\text{feat}}(\mathcal{F}(\hat{x}^{(l+1)}_{\text{2D}})), \\ A^{(l+1)}_{\text{2D}} &\leftarrow m^{(l+1)} \odot \phi_{\text{pos}}(\hat{x}^{(l+1)}_{\text{2D}}), \end{aligned} \tag{3}$$

$$A^{(l+1)}_{\text{3D}} \leftarrow \psi_{\text{pos}}(\bar{x}^{(l+1)}_{\text{3D}}). \tag{4}$$

Here, $m^{(l+1)}$ selects in-frame projections with valid positive depth, $\mathcal{F}(\hat{x}^{(l+1)}_{\text{2D}})$ denotes masked bilinear samples, and $\phi_{\text{feat}}, \phi_{\text{pos}}, \psi_{\text{pos}}$ are learned mappings. Eqs. (3)–(4) follow Eq. (2), producing

$(\mathcal{Q}^{(l+1)}, A^{(l+1)})$ for the next block; 3D token contents and unlisted components remain unchanged. Fig. 2 omits these updates. The sixth block skips feedback and generates final predictions from its parameter and box groups; "params" includes SMAL+ and camera parameters. Algorithm 1 in the Appendix details the complete inference forward pass.

Bipartite Matching for Multi-Animal Instances. To enable end-to-end training without heuristic post-processing such as Non-Maximum Suppression (NMS), we adopt a set prediction formulation following the DETR paradigm [12]. Specifically, we employ bipartite matching via the Hungarian algorithm [23] to find the optimal one-to-one assignment between the fixed-size set of P predicted animal hypotheses and the M ground-truth instances. The matching cost is a weighted combination of bounding box $\mathcal{L}_1$ distance, Generalized IoU [42], focal-style confidence penalty [44], and masked 2D keypoint distance. Once the optimal assignment is established, predicted outputs are reordered for loss computation. See Appendix for more details.

3.3 Loss Functions

After establishing the correspondence between predicted outputs and ground-truth labels via bipartite matching, we optimize the model using a multi-task loss function formulated as:

$$\mathcal{L} = \lambda_{\text{params}}\mathcal{L}_{\text{params}} + \lambda_{\text{2D}}\mathcal{L}_{\text{2D}} + \lambda_{\text{3D}}\mathcal{L}_{\text{3D}} + \lambda_{\text{box}}\mathcal{L}_{\text{box}}, \tag{5}$$

where $\lambda_{\{\cdot\}}$ denotes the weighting coefficients used to balance the respective loss contributions.

Parameter Loss ($\mathcal{L}_{\text{params}}$) computes the ℓ_2 distance between the predicted SMAL+ shape and pose parameters and their corresponding ground-truth values if provided.

Keypoint Losses ($\mathcal{L}_{\text{2D}}$, $\mathcal{L}_{\text{3D}}$) represent the ℓ_1 distance between the ground-truth 2D and 3D keypoint positions and the ones regressed from predicted SMAL+, respectively.

Bounding Box Loss ($\mathcal{L}_{\text{box}}$) supervises localization accuracy through a combination of coordinate regression, geometric alignment, confidence scoring and the denoising training strategy of DN-DETR [28]:

$$\mathcal{L}_{\text{box}} = \mathcal{L}_{\text{coord}} + \mathcal{L}_{\text{giou}} + \mathcal{L}_{\text{conf}} + \mathcal{L}_{\text{dn}}, \tag{6}$$

where $\mathcal{L}_{\text{coord}}$ is an ℓ_1 loss over normalized bounding box coordinates, and $\mathcal{L}_{\text{giou}}$ refers to the Generalized IoU (GIoU) loss [42]. To refine the objectness score, $\mathcal{L}_{\text{conf}}$ employs a Binary Cross-Entropy (BCE) loss, where the actual IoU between the matched predicted and ground-truth boxes serves as the soft target for the predicted confidence.

4 Herd3D Dataset



Figure 3: Example from the Herd3D dataset. This figure shows a generated scene with eight dogs, together with the corresponding canny map, depth map, RGB image, and 3D reconstruction overlay. The full Herd3D dataset contains multi-animal scenes with 2 to 8 animals per image.

To support multi-animal 3D reconstruction in real-world scenarios, we construct Herd3D, a large-scale dataset specifically designed for multi-animal scenes which contains over 5K images and 118 species (see Fig. 3). We believe that GenZoo [39] provides a strong and practical starting point for constructing large-scale animal datasets, because it couples a parametric animal model with controllable image synthesis, which enables scalable generation of paired images and geometry while maintaining pose and shape consistency. Building on GenZoo, we adapt the pipeline for multi-animal data generation. To construct group layouts, we sample up to 8 animals per image and place them on a shared ground plane. For each instance, we set $t_y = 0$ and sample translations with $t_x \in [-1.5, 1.5]$ using non-adjacent horizontal bins to limit excessive overlap, and $t_z \in [8, 50]$ from

predefined depth intervals while constraining the group depth span to at most 30; we add a small x and z jitter within ± 1.5 and apply a fixed ground alignment offset of 0.3. We further diversify global orientation by sampling pitch in $[-15^\circ, 15^\circ]$ and yaw in $[0^\circ, 360^\circ]$. To accommodate the increased complexity of multi-animal scenes-including frequent occlusions, higher ambiguity in instance-wise orientation and limited pose diversity, we adapt the GenZoo pipeline with several targeted modifications. We (i) impose scene layout constraints by placing all animals on a shared ground plane, (ii) expand the pose pool by integrating Animal3D poses [53] to increase pose diversity, (iii) replace the ControlNet backend with Qwen-Image-ControlNet-Union [49] to better preserve geometry and occlusion ordering, and (iv) resolve multi-animal orientation ambiguity via a two-stage Qwen3-VL-8B-Instruct [45] prompting scheme, which first predicts left-to-right per-animal facing directions and then composes a single coherent final prompt that integrates the species information, camera settings, and scene attributes. Each synthetic image has a resolution of 1024 × 1024 and includes annotations for SMAL+ parameters, 2D keypoints, 3D keypoints and bounding boxes.

5 Experiments

Datasets. We curate a comprehensive training corpus of 49.2K images containing both 2D and 3D annotations. Specifically, we aggregate the training splits of Animal Pose [10], APTv2 [57], AwA-Pose [5], Stanford Extra [7], Animal3D [53], and our newly introduced Herd3D. For evaluation, following the protocol in AniMer [36], we report results on two in-domain datasets (Animal3D and APTv2) alongside an out-of-domain (OOD) dataset Animal Kingdom.

Baselines. We benchmark our approach against three recent state-of-the-art (SOTA) methods to ensure a comprehensive evaluation across different architectural paradigms. For model-based techniques, we compare with AniMer [36], a transformer-based architecture utilizing the SMAL model, and GenZoo [39], which builds upon the SMAL+ variant. Additionally, we include 3D Fauna [30] as a representative SOTA model-free reconstruction approach.

Evaluation Metrics. We evaluate 3D accuracy using the Procrustes-Aligned Mean Per Joint Position Error (PA-MPJPE). For 2D accuracy, we report the Percentage of Correct Keypoints (PCK) and keypoint AP/mAP based on Object Keypoint Similarity (OKS) [31]. OKS is computed over visible keypoints, with errors normalized by the ground-truth visible-keypoint bounding-box area and $\sigma = 0.05$. We report AP^{kp}_{50} and AP^{kp}_{75} at OKS thresholds of 0.50 and 0.75, and $\text{mAP}^{kp}_{50:95}$ averaged over thresholds from 0.50 to 0.95 in steps of 0.05.

Implementation Details. Our network is optimized using AdamW [33] with an initial learning rate of 2×10^{-5}, incorporating a linear warmup over the first 15 epochs. Similar to AniMer [36], we employ a two-stage training strategy consisting of 250 epochs for the first stage and 250 epochs for the second. We apply prompt dropout for robustness: the mask prompt is dropped with 50% probability, the entire keypoint prompt is dropped with a probability of $p_{\text{drop}} = 0.2$, and otherwise each keypoint is independently masked out with a rate sampled uniformly from $[0,\ 0.7]$ per step, encouraging the model to handle partial or absent prompts at inference time. Training is distributed across four RTX 4090 GPUs with a gradient accumulation step of 16. To balance the objective function, our empirical loss weighting factors are set to $\lambda_{\text{params}} = 1$, $\lambda_{\text{2D}} = 5$, $\lambda_{\text{3D}} = 5$, and $\lambda_{\text{box}} = 1$.

5.1 Comparison

Comparison without prompts. We present the quantitative results in Table 1. Without any prompt, our method already achieves competitive or superior performance relative to existing approaches. On Animal3D, the prompt-free variant attains a PA-MPJPE of 80.7 mm, slightly outperforming AniMer (81.0 mm), while achieving a higher mAP (49.3 vs. 47.2). On APTv2, keypoint localization improves substantially, with PCK@0.1 reaching 87.9, far surpassing GenZoo (64.1) and AniMer (62.4), though AP remains lower (49.4 vs. 55.5 for GenZoo), indicating that the two paradigms exhibit complementary strengths under different metrics. On the OOD Animal Kingdom benchmark, our prompt-free results lead all metrics, demonstrating stronger generalization to unseen scenes.

Prompt-driven performance gains. A key advantage of our framework is its ability to leverage auxiliary prompts at inference time. When supplied with keypoints from an off-the-shelf ViTPose [54] estimator, performance improves consistently: on APTv2, AP rises from 49.4 to 55.5 and mAP from 23.5 to 27.9; on Animal Kingdom, AP increases from 45.0 to 50.5. This practical variant already

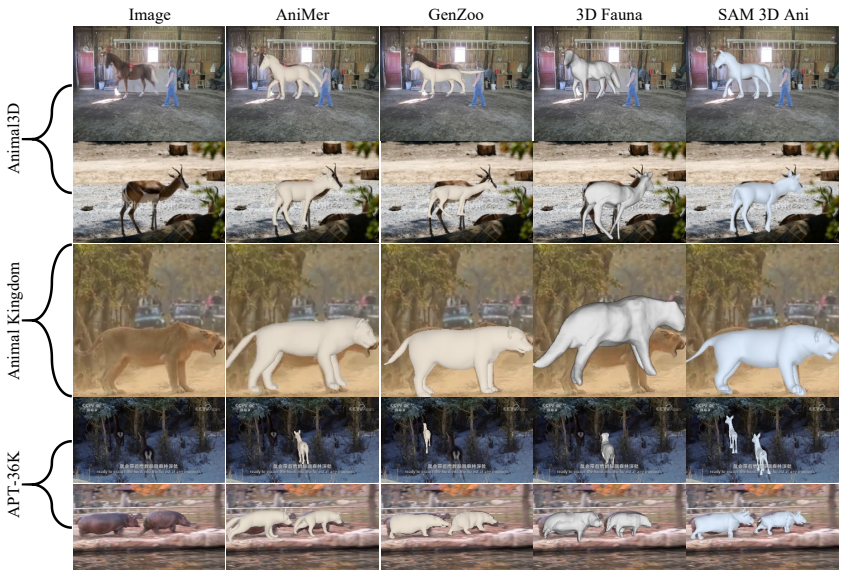


Figure 4: Qualitative comparisons on Animal3D, Animal Kingdom and APT-36K datasets. We compare our results with AniMer [36], GenZoo [39] and 3D Fauna [30]. AniMer, GenZoo, and 3D Fauna require pre-cropped single-animal images as input, whereas our method directly processes the full image and retrieves reconstructions whose confidence exceeds a predefined threshold, eliminating the need for cropping.

Table 1: **Quantitative comparisons on the Animal3D dataset, APTv2 dataset and Animal Kingdom dataset.**

Method	Animal3D			APTv2			Animal Kingdom		
	PA-MPJPE↓	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑
3D Fauna [30]	202.8	63.0	10.0	32.6	39.5	7.0	31.4	13.5	2.9
GenZoo [39]	123.2	85.3	34.0	64.1	55.5	19.2	42.1	23.6	4.7
AniMer [36]	81.0	93.4	47.2	62.4	51.8	20.1	58.2	39.7	10.4
Ours	80.7	92.8	49.3	87.9	49.4	23.5	59.2	45.0	12.6
Ours (w/ vitpose)	80.7	92.8	50.0	88.8	55.5	27.9	63.0	50.5	14.5
Ours (w/ GT prompt)	**75.8**	**93.8**	**53.4**	**89.0**	**57.4**	**29.7**	**66.2**	**60.1**	**17.7**

matches or surpasses the best baseline on most metrics. With ground-truth keypoint prompts, the gains become substantially larger: on APTv2, PCK@0.1 reaches 89.0 (vs. 62.4 for AniMer) and AP reaches 57.4 (vs. 55.5 for GenZoo); on Animal Kingdom, AP improves to 60.1 and mAP to 17.7, roughly doubling AniMer's 10.4. These results confirm that prompting provides a scalable mechanism for improving reconstruction quality, with performance increasing monotonically as prompt fidelity improves—a unique advantage that existing methods cannot replicate.

Qualitative comparison. Fig. 4 presents visual comparisons across the three benchmarks. 3D Fauna, as a model-free approach, produces coarse reconstructions that lack geometric detail. GenZoo and AniMer yield plausible shapes but exhibit less accurate alignment with the input image. Our method consistently produces reconstructions that are better aligned with the observed pose and viewpoint. Additional results spanning diverse species and challenging in-the-wild scenarios are shown in Fig. 5.

Performance under Varying Numbers of Animals. We further evaluate how reconstruction performance changes with the number of animals in an image. We stratify the APTv2 test set into images containing 1, 2, 3, 4, and 5+ annotated animals, corresponding to 3,502, 1,384, 557, 166, and 426 images, respectively (6,035 images in total). This grouping captures increasingly challenging conditions with stronger crowding, occlusion, and smaller animal scale. We report instance-averaged PCK@0.1, AP, and mAP for each group in Table 2. For direct comparison with existing methods, *Ours (w/ bbox)* follows the same protocol as the baselines: each animal is cropped using its ground-truth bounding box and reconstructed independently. Under this setting, SAM 3D Animal achieves the best performance across all metrics and animal-count groups. For example, it

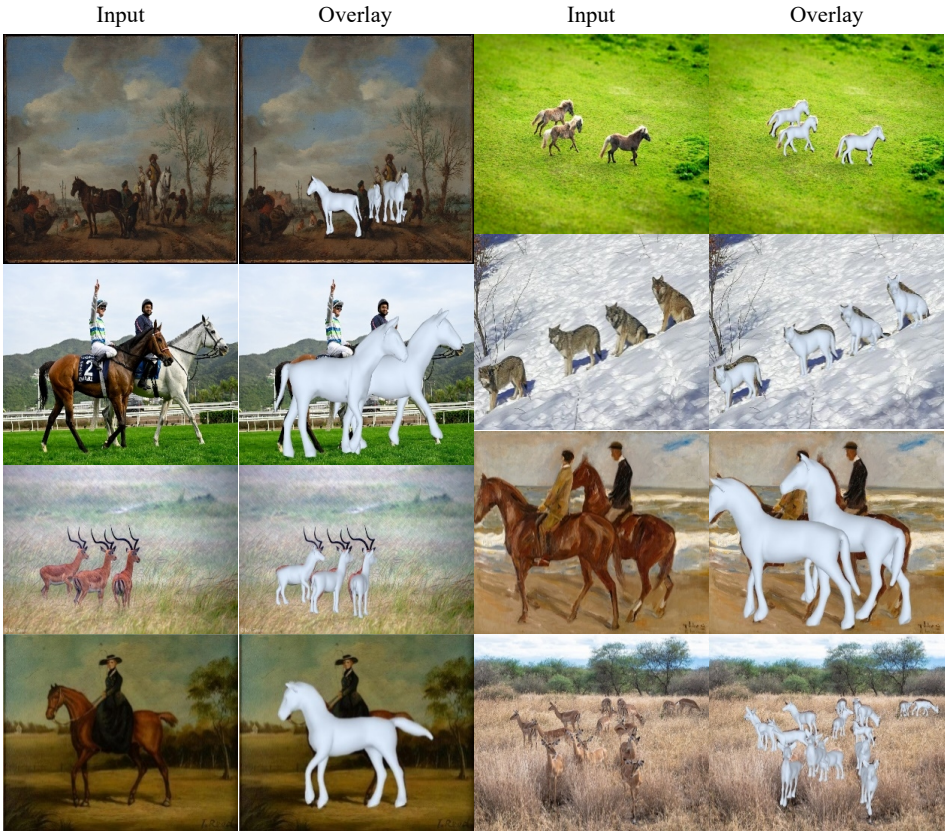


Figure 5: Qualitative evaluation of SAM 3D Animal. For each example, we show: (a) the input image, and (b) the 3D reconstruction overlay. To demonstrate the robustness of SAM 3D Animal, we visualize results across diverse animal species and challenging in-the-wild scenarios, including unusual poses, large viewpoint variations, and crowded scenes with several animals.

Table 2: **Quantitative comparisons on the APTv2 dataset stratified by the number of animals per image.**

Method	1 Animal			2 Animals			3 Animals			4 Animals			5+ Animals		
	PCK@0.1↑	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑
3D Fauna [30]	39.8	44.6	8.4	31.5	40.4	8.0	32.5	38.1	7.8	28.7	35.6	5.7	26.8	34.8	4.5
GenZoo [39]	68.6	57.5	19.8	58.7	39.3	12.0	59.7	40.4	12.7	57.5	45.6	15.2	48.9	24.1	6.6
AniMer [36]	78.4	69.0	29.8	64.2	53.1	19.6	61.8	55.9	20.9	52.0	50.2	19.2	46.7	31.5	10.4
Ours (w/o GT prompt)	83.4	81.1	42.9	64.9	50.4	21.8	62.1	48.0	21.2	42.7	31.9	12.7	28.9	13.3	4.0
Ours (w/ bbox)	**86.8**	**84.1**	**46.8**	**74.5**	**63.2**	**30.0**	**78.2**	**66.5**	**33.6**	**66.9**	**55.0**	**26.0**	**57.7**	**37.8**	**14.9**

obtains 86.8/84.1/46.8 PCK/AP/mAP on single-animal images and remains strongest for images with 5+ animals, reaching 57.7/37.8/14.9 and outperforming the best baseline by 11.0, 6.3, and 4.5 points, respectively. Without GT prompts, *Ours (w/o GT prompt)* processes the full image and jointly reconstructs all animals without bounding boxes, keypoints, or masks. It remains competitive for single-animal and moderately crowded scenes, but performance decreases more noticeably for 4 and 5+ animals, where jointly detecting and reconstructing all instances becomes substantially more difficult under severe occlusion and scale variation.

5.2 Ablation

We ablate three design axes to isolate their respective contributions: training data, prompt modality, and prompt density. Results are reported in Tables 3 and 4, with qualitative examples in Fig. 6.

Effect of Herd3D. Removing Herd3D from the training set leads to a consistent performance drop across all three benchmarks, with the largest degradation observed on APTv2 (Table 3). This is

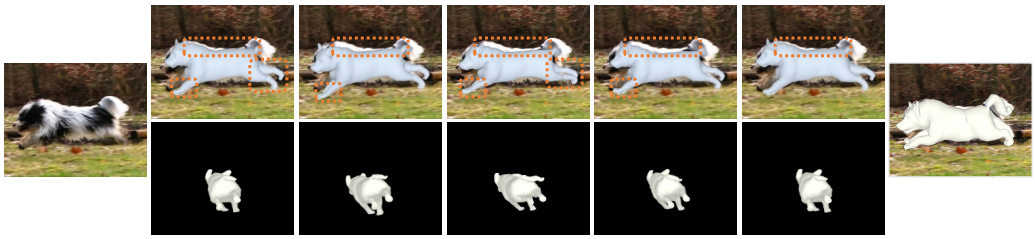


Figure 6: Ablation studies. Keypoint prompting, mask prompting, and training with our Herd3D dataset each lead to improved performance, as discussed in Sec. 5.2.

Table 3: **Ablation study on the Animal3D dataset, APTv2 dataset and Animal Kingdom dataset.**

Variant	Animal3D			APTv2			Animal Kingdom		
	PA-MPJPE↓	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑
w/o Herd3D	79.5	93.8	51.6	88.0	48.7	24.7	62.5	51.1	15.7
w/o prompt	80.7	92.8	49.3	87.9	49.4	23.5	59.2	45.0	12.6
w/o kp	80.8	93.1	49.8	87.9	49.7	23.5	59.3	45.4	12.8
w/o mask	**75.8**	**94.1**	**53.4**	88.9	57.3	29.6	66.0	**60.1**	17.5
Full	**75.8**	93.8	**53.4**	**89.0**	**57.4**	**29.7**	**66.2**	**60.1**	**17.7**

expected: Herd3D is the primary source of multi-animal scenes with per-instance annotations, and its absence disproportionately affects benchmarks that feature crowded or overlapping subjects. The result validates that our curated dataset fills a genuine gap in existing training resources rather than merely increasing data volume.

Keypoint vs. mask prompts. Comparing the prompt modality variants in Table 3 reveals a clear asymmetry: keypoint prompts are the dominant contributor, while mask prompts provide only marginal gains. Specifically, removing the mask prompt (*w/o mask*) leaves performance nearly unchanged from the full model, whereas removing keypoint prompts (*w/o kp*) reduces results to the level of the unprompted baseline. We attribute this to two factors. First, keypoints encode the articulated skeletal structure directly, which is precisely what the SMAL model needs to resolve pose ambiguity; masks, by contrast, convey only silhouette-level information that is largely redundant with the image features already extracted by the backbone. Second, the mask prompts are generated by SAM [11] at inference time, and segmentation errors, particularly on thin limbs or under occlusion, introducing noise that dilutes the prompt signal. Nonetheless, the full model still edges ahead of the keypoint-only variant on APTv2 and Animal Kingdom, suggesting that mask prompts offer a small but consistent complementary benefit when segmentation quality is adequate.

Number of keypoints. Table 4 displays the performance as a function of the number of keypoint prompts, ranging from 0 to 15. For each sample, we randomly select the specified number of keypoints from all available annotations as the prompt input. Performance improves monotonically, with the steepest gain occurring between 0 and 5 keypoints, even a sparse set of randomly chosen landmarks is sufficient to substantially disambiguate pose. Beyond 5, improvements continue at a diminishing rate, indicating that the initial keypoints resolve the most salient ambiguities while additional ones refine secondary joints. This graceful scaling is practically appealing: users can supply as few or as many keypoints as available, and the model extracts value from each additional annotation without requiring a fixed-size input.

5.3 Impact of Prompt Strategy Under Occlusion

To understand how different prompt strategies behave under varying levels of occlusion, we partition the test set by the number of visible keypoints into three groups (Low, Mid, High) and evaluate three prompt modes: GT Prompt (ground-truth 2D keypoints), ViTPose Prompt (automatically detected by ViTPose [54]), and No Prompt. Results are reported using mAP on both APTv2 and Animal Kingdom, as shown in Fig. 7. Note that for Animal Kingdom, the lowest visibility group starts at 8–11 keypoints because the dataset's annotations contain a minimum of 8 visible keypoints per instance.

Table 4: **Ablation study on the number of prompt keypoints.**

Number	Animal3D			APTv2			Animal Kingdom		
	PA-MPJPE↓	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑	PCK@0.1↑	AP↑	mAP↑
0	80.8	93.1	49.8	87.9	49.7	23.5	59.3	45.4	12.8
5	79.5	**94.1**	51.7	88.9	54.6	26.7	63.9	56.9	16.3
10	78.6	**94.1**	51.7	**89.1**	56.8	28.7	65.3	58.6	17.1
15	**77.4**	93.8	**52.5**	89.0	**57.5**	**29.6**	**66.0**	**60.2**	**17.7**

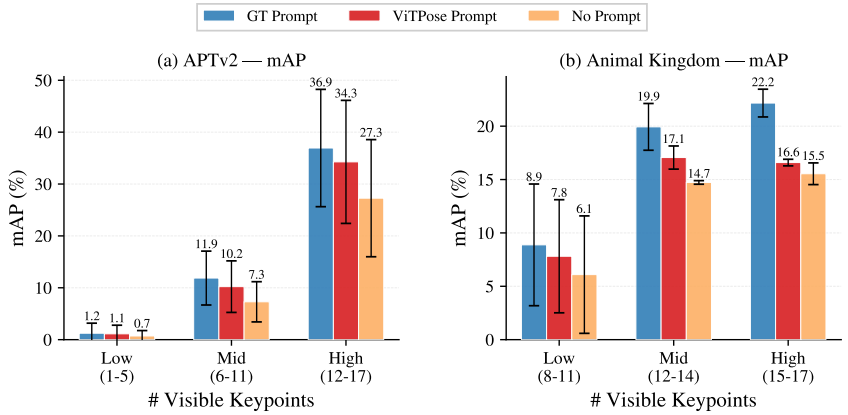


Figure 7: **Performance under different visibility levels.** We group test samples by the number of visible keypoints into Low, Mid, and High buckets. (a) mAP on APTv2. (b) mAP on Animal Kingdom. Error bars denote standard deviation across visibility counts within each group.

Prompt matters more under heavy occlusion. Across both datasets, the relative gain of prompting over the No Prompt baseline is largest in the Low visibility group and diminishes as more keypoints become visible. On APTv2, ViTPose Prompt improves mAP over No Prompt by 57% relative in the Low group, 40% in Mid, and 26% in High. On Animal Kingdom, the same trend holds: 28% relative gain in Low, 16% in Mid, and 7% in High. This confirms that the model relies more heavily on prompt-provided spatial priors when visual evidence is limited, and that prompting offers the greatest practical benefit precisely in the most challenging scenarios.

ViTPose Prompt is a practical alternative to GT. On APTv2, ViTPose Prompt consistently performs close to the GT upper bound across all visibility levels, demonstrating that an off-the-shelf keypoint detector can serve as an effective substitute for manual annotations at inference time.

6 Conclusion

We presented SAM 3D Animal, a promptable framework for multi-animal 3D reconstruction from a single image. Unlike prior animal reconstruction methods that predominantly focus on animal-centric images, our approach reconstructs multiple animals jointly through a set-prediction formulation and supports flexible keypoint and mask prompts to resolve ambiguity in crowded and occluded scenes.

Limitation. While SAM 3D Animal shows strong performance, it remains limited by the SMAL+ shape space and is therefore mainly applicable to quadruped-like animals. Moreover, relative depth ordering between animals is not explicitly constrained, which can cause inaccurate spatial arrangements under severe occlusion. Future work could explore more flexible animal representations, explicit depth-aware scene reasoning, and multi-view refinement by decoupling view-dependent and view-invariant parameters.

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A Bipartite Matching Details

We formulate the assignment between the predicted entities and ground-truth targets as a set prediction problem. Let $\hat{\mathcal{Y}} = \{\hat{y}_i\}_{i=1}^{N}$ denote the set of predictions, where each prediction comprises bounding box coordinates, a confidence score, and 2D keypoint locations. Let $\mathcal{Y} = \{y_j\}_{j=1}^{M}$ denote the set of ground-truth targets ($M \leq N$). Following DETR [12], the assignment is determined by searching for an optimal injection $\hat{\sigma} : \{1, \ldots, M\} \to \{1, \ldots, N\}$ that minimizes the overall bipartite matching cost:

$$\hat{\sigma} = \arg\min_{\sigma} \sum_{i=1}^{M} \mathcal{C}_{\text{match}}(\hat{y}_{\sigma(i)}, y_i) \tag{7}$$

where $\sigma(i) \in \{1, \ldots, N\}$ is the prediction index assigned to the i-th ground-truth target, and $\mathcal{C}_{\text{match}}(\hat{y}_{\sigma(i)}, y_i)$ is the pair-wise assignment cost between prediction $\sigma(i)$ and ground-truth target i. To ensure that the assigned predictions maintain high localization precision and structural pose alignment, our matching cost is a weighted composition of four distinct terms:

$$\mathcal{C}_{\text{match}} = \lambda_{\text{conf}}\mathcal{C}_{\text{conf}} + \lambda_{\text{bbox}}\mathcal{C}_{\text{bbox}} + \lambda_{\text{giou}}\mathcal{C}_{\text{giou}} + \lambda_{\text{kpts}}\mathcal{C}_{\text{kpts}} \tag{8}$$

Specifically, the weighting factors in the matching cost (Eq. 8) are set as follows: $\lambda_{\text{conf}} = 1$, $\lambda_{\text{bbox}} = 1$, $\lambda_{\text{giou}} = 1$, and $\lambda_{\text{kpts}} = 10$. The terms are defined as follows:

- **Bounding Box Cost ($\mathcal{C}_{\text{bbox}}$):** The $\mathcal{L}_1$ distance between the normalized center coordinates, width, and height of the predicted and ground-truth bounding boxes.
- **GIoU Cost ($\mathcal{C}_{\text{giou}}$):** The Generalized Intersection over Union (GIoU) cost [42], which provides a scale-invariant geometric alignment constraint for the bounding boxes.
- **Confidence Cost ($\mathcal{C}_{\text{conf}}$):** A focal-style penalty applied to the predicted confidence score $\hat{c}_i \in [0, 1]$. It is formulated as $\alpha(1 - \hat{c}_i)^{\gamma}(-\log \hat{c}_i)$ with focusing parameters $\alpha = 0.25$ and $\gamma = 2.0$ following [44], encouraging the model to match ground truth with high-confidence predictions.
- **Keypoint Cost ($\mathcal{C}_{\text{kpts}}$):** A pose-aware geometric constraint computed as the mean $\mathcal{L}_1$ distance between the predicted 2D keypoints and the corresponding ground-truth keypoints, strictly masked to account only for visible joints.

The optimal permutation $\hat{\sigma}$ is computed efficiently using the Hungarian algorithm [23]. Once the optimal assignment is established, the predicted outputs of multiple animals are reordered to perfectly align with the ground-truth sequence, allowing the subsequent calculation of the standard supervisory loss functions.

B More Details about Herd3D

We provide additional qualitative and statistical details of Herd3D in this section. Fig. 8 visualizes representative multi-animal examples with varying group sizes, where the number of visible animals gradually increases from two to eight. These examples illustrate the diversity of species, poses, viewpoints, background environment, and occlusion patterns covered by Herd3D, as well as the corresponding structural cues used for annotation and visualization, including edge maps, depth maps, RGB images, and pose overlays. Table 5 further summarizes the taxonomic composition of the training set by listing the top-10 animal families and representative species within each family. Together, these results show that Herd3D covers both dense multi-animal scenes and broad cross-species diversity, making it suitable for studying scene-centric animal pose understanding.

C Failure Cases in Herd3D Construction

During the construction of Herd3D, we observed several failure cases when rendering multi-animal scenes with Qwen-ControlNet. These failures are particularly common when animal meshes have large overlapping regions or severe inter-instance occlusions. In such cases, the rendered image may not faithfully preserve the intended 3D geometry and pose. For example, an animal whose head is oriented away from the camera may be incorrectly rendered with a forward-facing face, leading to inconsistent orientation between the mesh and the generated image. Local semantic errors may

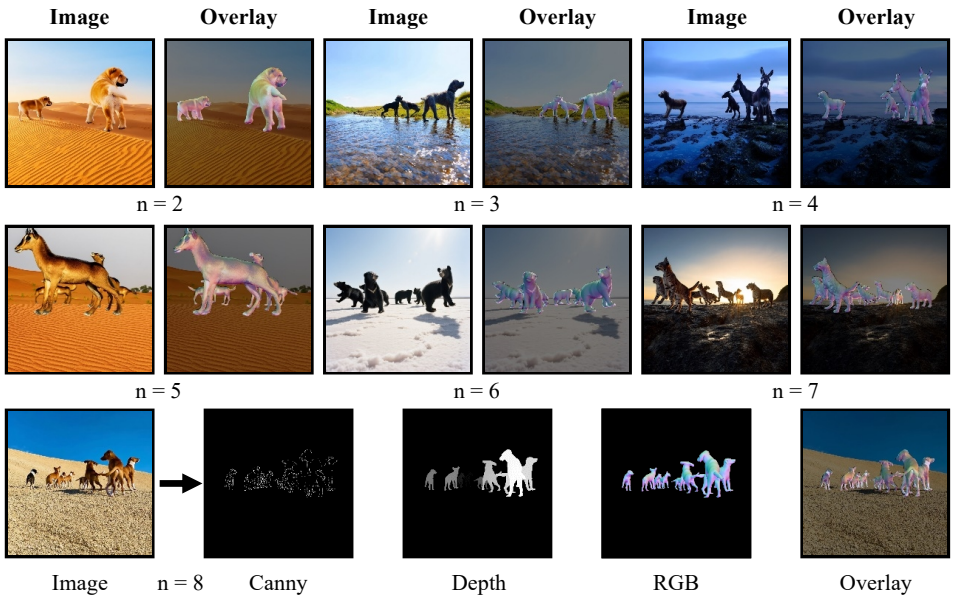


Figure 8: Herd3D multi-animal dataset. The images include dogs, horses, antelopes, bears, and cats, with the number of animals per image gradually increasing from 2 to 8 .



Figure 9: Failure cases of data generation.

also occur, where small body parts are misinterpreted, such as ears being rendered as noses or other facial structures. In addition, when two animals are spatially close, the renderer may blend their body regions, causing the torso or limbs of one animal to be partially rendered onto another. These artifacts indicate that dense multi-animal scenes remain challenging for image-conditioned generative rendering, especially under heavy occlusion and close physical interaction. We provide failure cases in Fig. 9.

Table 5: **Top-10 animal families in the Herd3D training image set.** Families are ranked by the number of labelled training images. For each family, we report representative species with their corresponding image counts.

Rank	Family	Images	Representative species
1	Canidae	3,238	Great Dane (105), Curly Coated Retriever (86), Irish Setter (85), Clumber Spaniel (84), Irish Wolfhound (84), Irish Water Spaniel (82), American Cocker Spaniel (73), American Eskimo Dog (71), Cairn Terrier (70), Alaskan Malamute (64)
2	Bovidae	324	Cow (106), Cattle (95), American Bison (21), Antelope (21), Goat (21), Bighorn Sheep (20), Buffalo (20), Impala (20)
3	Equidae	305	Horse (127), Feral Horse (115), Domestic Horse (21), Donkey (21), Common Zebra (21)
4	Mustelidae	187	Honey Badger (73), Eurasian Otter (54), African Clawless Otter (20), American Badger (20), American Mink (20)
5	Rhinocerotidae	157	Black Rhinoceros (157)
6	Ursidae	151	American Black Bear (90), Brown Bear (21), Da Xiong Mao (20), Giant Panda (20)
7	Felidae	143	Bay Lynx (22), European Wildcat (21), Bay Cat (20), Caracal (20), Cat (20), Cheetah (20), Cougar (20)
8	Giraffidae	63	Angolan Giraffe (21), Cape Giraffe (21), Giraffe (21)
9	Herpestidae	42	Gray Meerkat (42)
10	Viverridae	41	Bear Cat (21), African Civet (20)

Table 6: Forward-pass inference time under different numbers of animals on APTv2. The reported time is measured in milliseconds per image and excludes data loading, preprocessing, and metric computation.

Method	1 Animal	2 Animals	3 Animals	4 Animals	5+ Animals
3D Fauna	195.9 (195.9)	198.1 (396.2)	196.6 (589.9)	198.1 (792.6)	195.4 (1518.1)
GenZoo	7.8 (7.8)	7.8 (15.5)	8.0 (24.0)	8.0 (31.8)	7.8 (60.9)
AniMer	7.6 (7.6)	7.6 (15.1)	7.6 (22.7)	7.6 (30.3)	7.6 (59.5)
Ours	79.4	79.2	84.7	80.3	80.5

D More Ablation Experiments

Fig. 10 further demonstrates the effect on the number of prompt keypoints. With more keypoints used as prompt information, SAM 3D Animal produces results aligned better with input image.



Input Image | w/o kp | kp = 5 | kp = 10 | kp = 15

Figure 10: Ablation study on the number of prompt keypoints.

E Inference Efficiency

We additionally compare the inference efficiency of different methods on APTv2. We group test images by the number of annotated animals in the original image: 1, 2, 3, 4, and 5+ animals, containing 3,502, 1,384, 557, 166, and 426 images, respectively. Table 6 reports the average inference time for each group.

During measurement, we excluded data loading, preprocessing, and metric computation. All measurements were performed on a single NVIDIA RTX 4090 GPU. Note that, for 3D Fauna, GenZoo and AniMer, the input is cropped animal image resized to 256x256 resolution with padding. For SAM 3D Animal, the input is the whole uncropped image resized to 512x512 resolution with padding. Note that, although our method is currently slower than GenZoo and AniMer without any acceleration techniques, it maintains a stable time cost for different numbers of animals, thus having the potential to scale up for inferring a large number of animals efficiently.

F Robustness to Image Quality

We further study how input image quality affects reconstruction performance on Animal3D. We divide the test set into three approximately balanced groups using an object-centric image-quality score that equally combines percentile-normalized effective animal resolution and crop sharpness. The resulting groups are: *Low* ($[0, 0.3903]$, 107 images), *Medium* ($[0.3903, 0.6270]$, 106 images), and *High* ($[0.6301, 0.9483]$, 107 images). Table 7 reports PA-MPJPE, AP, and mAP for each group.

For direct comparison with the baselines, *Ours (w/ bbox)* follows the same evaluation protocol: each animal is cropped using its ground-truth bounding box and reconstructed independently. In contrast, *Ours (w/o GT prompt)* takes the full image as input and reconstructs the animal without ground-truth bounding boxes, keypoints, or masks.

Image quality primarily affects the 2D reconstruction metrics, while PA-MPJPE remains relatively stable. In the prompt-free setting, mAP increases from 35.9 on low-quality images to 59.7 on high-quality images. Bounding-box localization substantially improves robustness to degraded images: the low-to-high AP gap is reduced from 12.2 to 6.6 points, while the mAP gap decreases from 23.8 to 15.6 points. These results suggest that accurate localization can partially compensate for

Table 7: **Quantitative comparisons on the Animal3D dataset under different image-quality level.**

Method	Low			Medium			High		
	PA-MPJPE↓	AP↑	mAP↑	PA-MPJPE↓	AP↑	mAP↑	PA-MPJPE↓	AP↑	mAP↑
3D Fauna [30]	194.7	66.3	10.5	212.8	59.2	9.4	201.0	63.4	10.1
GenZoo [39]	124.8	80.4	29.8	118.6	88.7	35.5	124.3	85.0	37.3
AniMer [36]	**82.6**	86.3	40.8	78.1	98.5	48.8	82.3	95.5	47.2
Ours (w/o GT prompt)	84.3	85.0	35.9	79.5	96.2	52.5	78.4	97.2	59.7
Ours (w/ bbox)	83.2	**92.5**	**47.3**	**78.0**	**99.1**	**57.8**	**76.0**	**99.1**	**62.9**

Table 8: Effect of keypoint prompt locations on the 252-image common-visible subset of Animal3D. *Distal* uses the four fore- and hind-paw endpoints; *Intermediate* uses the four intermediate limb joints, approximately corresponding to the front elbows and rear knees/hocks; *Proximal* uses the four shoulder and hip keypoints; and *Random* samples four visible keypoints from all 26 Animal3D keypoints.

Prompt type	PA-MPJPE ↓	AP ↑	mAP ↑
No Prompt	77.6	95.6	51.4
Random	76.4	95.4	52.3
Distal	76.1	**96.4**	52.5
Intermediate	76.0	96.0	**54.0**
Proximal	**75.8**	95.6	53.4

limited visual detail, while the bbox-based setting maintains strong performance across all three image-quality levels.

G Effect of different types of keypoint prompts.

In Table 8, we conduct an additional controlled experiment on Animal3D to investigate whether the anatomical location of keypoint prompts affects reconstruction quality. To ensure a fair comparison, we select a common subset of 252 images in which all candidate keypoints are visible and provide exactly four keypoint prompts for each image. We compare Distal keypoints (the four fore- and hind-paw endpoints), Intermediate keypoints (the front elbows and rear knees/hocks), Proximal keypoints (the shoulder and hip keypoints), and Random keypoints (four visible keypoints sampled from all 26 Animal3D keypoints). The Random result is averaged over five seeds, and mask prompts are disabled to isolate the effect of keypoint location.

The results confirm that keypoint selection has a measurable effect. Anatomically selected prompts generally outperform random prompts. Proximal keypoints achieve the best PA-MPJPE of 75.8, suggesting that shoulder and hip landmarks provide stronger constraints for global 3D pose recovery. Intermediate keypoints achieve the best mAP of 54.0, indicating that articulated limb joints are more informative for fine-grained pose estimation. Distal keypoints achieve the best AP of 96.4. Therefore, distal keypoints are informative, as suggested by the reviewer, but they are not consistently optimal across all metrics. We include this controlled analysis and the corresponding discussion in the revised manuscript.

H Complete Inference Forward Pass

Algorithm 1 summarizes the prompt-conditioned forward pass using the indexing of Eqs. (1)–(4). Here, κ denotes image/camera metadata, $\hat{\Theta}$ collects SMAL+ parameters, and $\hat{C}$ denotes camera predictions.

Algorithm 1 SAM 3D Animal inference forward pass ($P = 30$, $L = 6$)

Require: Image $\boldsymbol{x}$, keypoint prompts $\mathcal{P}$, optional mask M, metadata κ, optional previous estimate η_{prev}
Ensure: P scored hypotheses with SMAL+ parameters, cameras, boxes, meshes, and keypoints

$\mathcal{F} \leftarrow \text{ImageFeatures}(\boldsymbol{x}, M, \kappa)$
$\eta_0 \leftarrow \text{LearnedInitialEstimate}(P)$
$\eta_{\text{prev}} \leftarrow \eta_0$ if no previous estimate is supplied
$(\mathcal{Q}^{(0)}_{\text{base}}, A^{(0)}_{\text{base}}) \leftarrow \text{InitBase}(\eta_0, \eta_{\text{prev}}, \kappa)$
$\mathcal{P} \leftarrow \text{PadMissingPrompts}(\mathcal{P}, c = -2)$
$\mathcal{Q}^{(0)}_{\text{prompt}} \leftarrow \text{Concat}_i\big(W_p\,\text{EncodePoint}(p_i) + b_p\big)$
$\mathcal{Q}^{(0)} \leftarrow \text{Concat}(\mathcal{Q}^{(0)}_{\text{base}}, \mathcal{Q}^{(0)}_{\text{prompt}})$ ▷ Inject prompts once
$A^{(0)} \leftarrow \text{Concat}(A^{(0)}_{\text{base}}, \mathcal{Q}^{(0)}_{\text{prompt}})$
for $l = 0, \ldots, L-1$ **do**
  $\mathcal{Q}^{(l+1)} \leftarrow \mathcal{D}_{l+1}(\mathcal{Q}^{(l)}, A^{(l)}, \mathcal{F})$ ▷ Eq. (2)
  $A^{(l+1)} \leftarrow A^{(l)}$
  $(\hat{\Theta}, \hat{C}, \hat{B}, \hat{s})^{(l+1)} \leftarrow \text{Predict}(\mathcal{Q}^{(l+1)}_{\text{params}}, \mathcal{Q}^{(l+1)}_{\text{box}}; \eta_0)$
  $(\mathbf{V}, \mathbf{X}_{\text{3D}})^{(l+1)} \leftarrow \mathcal{M}(\hat{\Theta}^{(l+1)})$
  $(\hat{x}_{\text{2D}}, d)^{(l+1)} \leftarrow \text{Project}(\mathbf{X}^{(l+1)}_{\text{3D}}, \hat{C}^{(l+1)}; \kappa)$
  if $l < L-1$ **then** ▷ Feedback only after the first five blocks
    $m^{(l+1)} \leftarrow \text{ValidProjection}(\hat{x}^{(l+1)}_{\text{2D}}, d^{(l+1)})$
    $\bar{x}^{(l+1)}_{\text{3D}} \leftarrow \text{CenterPelvis}(\mathbf{X}^{(l+1)}_{\text{3D}})$
    $f^{(l+1)} \leftarrow m^{(l+1)} \odot \text{BilinearSample}(\mathcal{F}, \hat{x}^{(l+1)}_{\text{2D}})$
    $\mathcal{Q}^{(l+1)}_{\text{2D}} \leftarrow \mathcal{Q}^{(l+1)}_{\text{2D}} + \phi_{\text{feat}}(f^{(l+1)})$
    $A^{(l+1)}_{\text{2D}} \leftarrow m^{(l+1)} \odot \phi_{\text{pos}}(\hat{x}^{(l+1)}_{\text{2D}})$ ▷ Eq. (3)
    $A^{(l+1)}_{\text{3D}} \leftarrow \psi_{\text{pos}}(\bar{x}^{(l+1)}_{\text{3D}})$ ▷ Eq. (4)
  end if
end for
return $\{\hat{\Theta}^{(L)}_j, \hat{C}^{(L)}_j, \hat{B}^{(L)}_j, \hat{s}^{(L)}_j, \mathbf{V}^{(L)}_j, \mathbf{X}^{(L)}_{\text{3D},j}, \hat{x}^{(L)}_{\text{2D},j}\}_{j=1}^{P}$

Initialization and prompting. ImageFeatures extracts ViT features with the configured mask and ray conditioning. InitBase uses learned parameters to construct the four base query groups and the previous-estimate conditioning tokens included in the parameter-related inputs. Their initial positional augmentations contain the previous-estimate embeddings, with zero augmentation for the box and keypoint groups. EncodePoint is the piecewise encoding defined in Section 3.2: valid keypoint IDs use position-plus-label embeddings, while labels -1 and -2 use the negative and invalid embeddings. Missing prompts are represented by invalid entries.

Predictions and feedback. Predict includes decoder-output normalization and the SMAL+, camera, box, and confidence heads. Project produces crop-image coordinates and depths; m selects in-frame projections with valid positive depth. The mask is applied before feature projection and after positional encoding, as in the implementation. Feedback uses the keypoint subsets associated with the 2D and 3D token groups, and preserves 3D token contents and all unlisted components. All hypotheses are returned with confidence scores; a configured confidence threshold can select output instances. During training, denoising queries, auxiliary supervision, and bipartite matching supplement this inference path as described in Section 3.3 and Appendix A.

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Justification: The paper introduces Herd3D and documents its generation process, scene layout constraints, animal placement strategy, pose sources, ControlNet backend, orientation prompting scheme, image resolution, and available annotations.

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Answer: [Yes]

Justification: The paper describes the use of Qwen3-VL-8B-Instruct in the Herd3D generation pipeline to resolve multi-animal orientation ambiguity, and also describes the use of Qwen-Image-ControlNet-Union as the image generation backend. These uses are part of the dataset generation method rather than only writing or editing assistance.

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